



Advanced PG Certificate in AI Engineering on Cloud and AIOps
Lesson Plan - IIT Roorkee | Cohort 1

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Dates	Days	# Session	Slot	Topics	Subtopics
14-Sep-25	Sunday	1	3:00 PM to 5:00 PM	Topic 1: Foundations of AI ML and Generative AI	Subtopic 1: Understanding Supervised, Unsupervised, and Reinforcement Learning, Data Manipulation for Machine Learning
20-Sep-25	Saturday	2	3:00 PM to 5:00 PM		Subtopic 2: Exploratory Data Analysis, Essential Math for ML & AI, Definition of Generative AI, Applications, and LLMs
21-Sep-25	Sunday	3	10:00 AM to 12:00 PM	Topic 2: Deep dive into Machine Learning Algorithms	Subtopic 1: Introduction to Regression in Supervised Learning
27-Sep-25	Saturday	4	10:00 AM to 12:00 PM		Subtopic 2: Linear Regression, Decision Tree Regressor - Concepts and Hands-On
28-Sep-25	Sunday	5	10:00 AM to 12:00 PM		Subtopic 3: Random Forest Regressor, Regularization Techniques and Regression Evaluation Metrics
04-Oct-25	Saturday	6	10:00 AM to 12:00 PM		Subtopic 4: Logistic Regression Classifier, Decision Tree Classifier - Concepts and Hands-On
05-Oct-25	Sunday	7	05:00 PM to 7:00 PM		Subtopic 5: Random Forest Classifier, Dimensionality Reduction and Classification Evaluation Metrics
11-Oct-25	Saturday	8	10:00 AM to 12:00 PM		Subtopic 6: Unsupervised Machine Learning - KMeans Clustering and others
12-Oct-25	Sunday	9	8:30 AM to 10:30 AM		Subtopic 7: Reinforcement Machine Learning - 1

18-Oct-25	Saturday	10	8:30 AM to 10:30 AM		Subtopic 8: Reinforcement Machine Learning - 2
19-Oct-25	Sunday	11	8:30 AM to 10:30 AM		Subtopic 9: Reinforcement Machine Learning - 3
25-Oct-25	Saturday	12	3:00 PM to 5:00 PM	Topic 3: Understanding Deep Learning	Subtopic 1: Exploring Deep Learning and ANN
26-Oct-25	Sunday	13	3:00 PM to 5:00 PM		Subtopic 2: Neural Networks with TensorFlow (Basics + Hands-on)
1-Nov-25	Saturday	14	3:00 PM to 5:00 PM		Subtopic 3: CNNs and YOLO for Object Detection
2-Nov-25	Sunday	15	3:00 PM to 5:00 PM		Subtopic 4: Transfer Learning and RNNs (LSTM)
8-Nov-25	Saturday	16	3:00 PM to 5:00 PM	Topic 4: Exploring NLP	Subtopic 1: Understanding Text for NLP + Tokenization, Stemming, Lemmatization
9-Nov-25	Sunday	17	3:00 PM to 5:00 PM		Subtopic 2: Stopwords, Keywords, Text Vectorization (TF-IDF, BoW, Word2Vec)
15-Nov-25	Saturday	18	3:00 PM to 5:00 PM		Subtopic 3: RNN Architectures & NLP Use Cases across domains
16-Nov-25	Sunday	19	8:30 AM to 10:30 AM	Topic 5: Exploring Cloud Ecosystems	Subtopic 1: Introduction to Cloud Ecosystem, Definitions, Cloud Characteristics
22-Nov-25	Saturday	20	8:30 AM to 10:30 AM		Subtopic 2: Deployment Models, Leading Cloud Providers, AWS vs Azure vs GCP

23-Nov-25	Sunday	21	8:30 AM to 10:30 AM		Subtopic 3: Data Centers, SaaS/IaaS/PaaS, and Cloud Issues & Challenges
29-Nov-25	Saturday	22	TBD	MasterClass	
30-Nov-25	Sunday	23	TBD	MasterClass	
6-Dec-25	Saturday	24	8:30 AM to 10:30 AM	Topic 6: Advancement in Hypervisors	Subtopic 1: Introduction to Hypervisors, Virtualization Basics, Reference Models
7-Dec-25	Sunday	25	8:30 AM to 10:30 AM		Subtopic 2: Virtualization Characteristics and Hypervisor Interfaces
13-Dec-25	Saturday	26	8:30 AM to 10:30 AM		Subtopic 3: Type-1 vs Type-2 Hypervisors and Design Differences
14-Dec-25	Sunday	27	8:30 AM to 10:30 AM		Subtopic 4: Virtualization Methods: Full, Para, Hardware-Assisted
20-Dec-25	Saturday	28	8:30 AM to 10:30 AM		Subtopic 5: Memory, I/O, and OS Virtualization
21-Dec-25	Sunday	29	8:30 AM to 10:30 AM		Subtopic 6: Performance Analysis, Bottlenecks, and Optimization
27-Dec-25	Saturday	30	3:00 PM to 5:00 PM	Topic 7: Engaging with Generative AI	Subtopic 1: How Generative AI works, Text/Image/Audio/Video Generation
28-Dec-25	Sunday	31	3:00 PM to 5:00 PM		Subtopic 2: Using Hugging Face for Text & Image Models, GANs and Types of GANs (ProGAN, SRGAN, CycleGAN)

3-Jan-26	Saturday	32	3:00 PM to 5:00 PM		Subtopic 3: Autoencoders, Variational Autoencoders (VAEs), Diffusion Models
4-Jan-26	Sunday	33	TBD	MasterClass	
10-Jan-26	Saturday	34	3:00 PM to 5:00 PM	Topic 8: Understanding LLMs	Subtopic 1: Prompt Tuning Techniques
11-Jan-26	Sunday	35	3:00 PM to 5:00 PM		Subtopic 2: Attention Mechanism and Transformer Architecture
17-Jan-26	Saturday	36	3:00 PM to 5:00 PM		Subtopic 3: Encoder-Decoder Architectures + Text Generation
18-Jan-26	Sunday	37	3:00 PM to 5:00 PM		Subtopic 4: BERT for Transfer Learning + Pretrained LLMs from Hugging Face
24-Jan-26	Saturday	38	3:00 PM to 5:00 PM		Subtopic 5: Fine-tuning LLMs
25-Jan-26	Sunday	39	3:00 PM to 5:00 PM	Topic 9: GenAI Application Development Prompting Techniques	Subtopic 1: Prompt Engineering Basics, Prompt Types, Examples of Good vs Bad Prompts
31-Jan-26	Saturday	40	3:00 PM to 5:00 PM		Subtopic 2: OpenAI, GPT, Playground, API Cost/Latency, Multimodal Prompting, DALL-E-3
01-Feb-26	Sunday	41	3:00 PM to 5:00 PM		Subtopic 3: Prompt Evaluation, LangChain Agents, Tool Integration, Zero-shot/Conversational Agents
07-Feb-26	Saturday	42	3:00 PM to 5:00 PM	Topic 10: Synthetic Data and Datasets for LLMs	Subtopic 1: Introduction to Synthetic Data, Generating Synthetic Data, Synthetic Data for LLMs, Real-world Applications and Use Cases, Hands-on generating and using Synthetic Data

8-Feb-26	Sunday	43	3:00 PM to 5:00 PM	Topic 11: AI Embeddings & Retrieval	Subtopic 1: Understanding AI Embeddings, Hugging Face Embeddings
14-Feb-26	Saturday	44	3:00 PM to 5:00 PM		Subtopic 2: Vector Databases in AI, CRUD Operations, RAG and Retrieval-Augmented Generation
15-Feb-26	Sunday	45	3:00 PM to 5:00 PM		Subtopic 3: RAG solutions using OpenAI and Hugging Face Models, Real-world Applications
21-Feb-26	Saturday	46	3:00 PM to 5:00 PM		Subtopic 4: Ethical Considerations, Hallucinations, Drift, Bias, Security & Privacy in Embeddings
22-Feb-26	Sunday	47	TBD	MasterClass	
28-Feb-26	Saturday	48	3:00 PM to 5:00 PM	Topic 12: Understanding Agentic AI	Subtopic 1: Intro to Agentic AI, Agent Definition & Autonomy
1-Mar-26	Sunday	49	3:00 PM to 5:00 PM		Subtopic 2: Simple vs. Knowledge-Based Agents, Reflex vs. Goal-Driven Agents
07-Mar-26	Saturday	50	3:00 PM to 5:00 PM		Subtopic 3: Agents with - Lang Graph, SmolAgents, AutoGen and other Frameworks
08-Mar-26	Sunday	51	3:00 PM to 5:00 PM		Subtopic 4: Integrating Knowledge Bases (RAG, Domain Data), Measuring Performance
14-Mar-26	Saturday	52	3:00 PM to 5:00 PM		Subtopic 5: Hierarchical Agent Planning, Multi-Step Reasoning with LLMs
15-Mar-26	Sunday	53	3:00 PM to 5:00 PM		Subtopic 6: Memory, Retrieval Augmentation & Domain-Specific Prompting

21-Mar-26	Saturday	54	8:30 AM to 10:30 AM	Topic 13: Container Orchestration	Subtopic 1: Understanding LLM Deployment Architectures + FastAPI + Docker
22-Mar-26	Sunday	55	8:30 AM to 10:30 AM		Subtopic 2: Managing GPU Workloads, Scaling with Kubernetes & Istio
28-Mar-26	Saturday	56	8:30 AM to 10:30 AM		Subtopic 3: Optimizing Latency and Throughput for LLM Containers
29-Mar-26	Sunday	57	8:30 AM to 10:30 AM		Subtopic 4: Secure Access, Rate Limiting for AI APIs, CI/CD for Microservices
04-Apr-26	Saturday	58	8:30 AM to 10:30 AM		Subtopic 5: Monitoring, Logging, Model Versioning
05-Apr-26	Sunday	59	8:30 AM to 10:30 AM		Subtopic 6: Rollbacks & Cost Optimization Strategies for LLM Inference
11-Apr-26	Saturday	60	3:00 PM to 5:00 PM	Topic 14: Cloud Security & Resilience	Subtopic 1: Network, Host, and Application-Level Security
12-Apr-26	Sunday	61	3:00 PM to 5:00 PM		Subtopic 2: Data Security and Lifecycle Management
18-Apr-26	Saturday	62	3:00 PM to 5:00 PM		Subtopic 3: Identity and Access Management (IAM) and API Security
19-Apr-26	Sunday	63	3:00 PM to 5:00 PM		Subtopic 4: Regulatory Compliance, Governance and Zero-Trust Architecture
25-Apr-26	Saturday	64	3:00 PM to 5:00 PM		Subtopic 5: AI model security and Operational Security

26-Apr-26	Sunday	65	3:00 PM to 5:00 PM		Subtopic 6: Risk Management, Business and Legal Consideration
2-May-26	Saturday	66	8:30 AM to 10:30 AM	Topic 15: Understanding Cloud in context of Gen AI and LLMs	Subtopic 1: EC2 Deep Dive, AMIs, and AWS Storage Solutions (EBS, S3, EFS)
3-May-26	Sunday	67	8:30 AM to 10:30 AM		Subtopic 2: Load Balancing, Auto Scaling, and AWS Lambda
9-May-26	Saturday	68	8:30 AM to 10:30 AM		Subtopic 3: AWS Bedrock for GenAI and Deploying Open-Source LLMs on EC2/EKS
10-May-26	Sunday	69	8:30 AM to 10:30 AM		Subtopic 4: Fine-Tuning and Inference Pipelines on Cloud
16-May-26	Saturday	70	8:30 AM to 10:30 AM		Subtopic 5: Cost, Performance Optimization, and SageMaker Pipelines
17-May-26	Sunday	71	8:30 AM to 10:30 AM		Subtopic 6: Model Versioning, A/B Testing, and Security & Compliance in Production
Note: The schedule is tentative and subject to change in case of unforeseen circumstances.					